Bavan P

sm25mtech11009@iith.ac.in | bavansp@gmail.com | +91 7010894009 | LinkedIn | GitHub

EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	7.46
Notre dame of holy cross	Class XII (CBSE)	2021	90.00%
Notre dame of holy cross	Class X (CBSE)	2019	87.60%

EXPERIENCE

Graduate Student Researcher, TiHAN -- IIT Hyderabad

- **Spearheaded a 5-member team** at TiHAN to retrofit an ICE vehicle into an EV, managing component procurement, technical task delegation, and weekly milestone reporting to the Project Director.
- **Demonstrated rapid cross-disciplinary adaptability**, stepping beyond core domains to troubleshoot mechanical integration, high-voltage wiring, and low-level firmware to achieve full vehicle operational bring-up.
- **Rapidly prototyped and validated custom AV features**, moving ideas from concept to live vehicle testing.

Robotics Software Intern, Crobot Technologies

- **Drove end-to-end UBOT telepresence robot development** and built custom GUI for teleoperation.
- **Deployed autonomous navigation systems** at Cyient (Hyderabad), achieving 100% uptime for **over a year**.
- **Architected embedded stack** and executed **low-level bring-up** across Jetson and Raspberry Pi SBCs.

PROJECTS

Autonomous Robotaxi Retrofitting & Control

- **Engineered mechanical/electrical EV retrofitting** to establish a campus autonomous robotaxi platform.
- **Developed ROS Noetic/Python control stack** with Kinematic MPC, verified in high-fidelity CARLA simulations.
- **Integrated real-world vehicle dynamics** into core control modules to build a field-ready vehicle package.
- **Benchmarked customised Pure Pursuit and stanley controller against Kinematic MPC**, demonstrating superior trajectory tracking.
- **Implemented point-to-point path planning**, computing optimal shortest routes between user-selected coordinates.

Transformer-Based Multimodal Trajectory Planning (CARLA)

- Developing a **hybrid autonomy architecture** combining a **generative trajectory planner** with **MPC** for **dynamic obstacle avoidance**, encoding the **past 20 ego-states** for smooth, **low-jerk motion**.
- Integrated **YOLO 2D detections** into the **Transformer planner** to predict dynamic agent intent and path options.
- Coupling **Transformer proposals** with **kinematic MPC** to select **collision-free** paths within steering and acceleration limits.
- Benchmarked **multimodal sampling**, generating diverse candidate paths for overtaking, following, and yielding.

LiDAR-Guided 3D Gaussian Splatting (DrivingScene-3DGS)

- **Extracted drift-free LiDAR priors** at TiHAN to establish ground-truth maps and trajectories.
- **Architected LiDAR-Prior 3DGS pipeline**, seeding 1M Gaussians to eliminate VIO ghosting artifacts.
- **Calibrated 6 surround vehicle cameras**, optimizing spherical harmonics and opacity for view synthesis.
- **Utilized colored point clouds for drivable area segmentation**, extracting road boundaries for global path planning.
- **Implemented photo-goal dispatch**, allowing users to select destination images to navigate the vehicle via Interactive A*.

Indoor Autonomous Telepresence Robot (UBOT)

- **Co-developed ROS 2 telepresence robot** in a 3-member team, designing full software stack and custom GUI.
- **Engineered differential-drive base** fusing dual LiDARs, IMU, and RealSense to eliminate blind spots.

- **Deployed ROS 2 Nav2 stack** with MPPI control and SLAM Toolbox for dynamic, collision-free navigation.
- **Configured Nav2 behavior trees** with custom recovery routines to resolve deadlocks autonomously.
- **Engineered dual-LiDAR scan fusion** to eliminate blind spots, deploying dynamic free-space maze navigation.

Medical Insurance Cost Analytics & Predictive Modeling

- **Built automated PostgreSQL ETL pipeline** using psycopg2 and SQLAlchemy for structured demographic ingestion.
- **Analyzed demographics** via SQL and Seaborn/Matplotlib, isolating smoking as driving 60% cost variance.
- **Trained Random Forest Regressor** using scikit-learn, achieving **>86% R2 accuracy** for charge prediction.

AI-Powered Game Solvers & Move Predictor (Sudoku & Chess)

- **Engineered dual-solver in PyTorch** combining 9-layer **SudokuCNN** with exact recursive backtracking.
- **Designed 3x3 convolutional kernels** for relational constraint propagation and 15 ms board inference.
- **Benchmarked Deep Learning vs RL self-play** for Chess move prediction to evaluate decision efficiency.

STM32-Based Educational Robot ("Tulip")

- **Programmed STM32 (ARM Cortex-M) in Embedded C** to control DC motors, OLED display, and power circuits.
- **Integrated ultrasonic distance sensors and servos** to build automated obstacle-scanning routines.
- **Configured UART Bluetooth telemetry** to interface with a mobile application for real-time teleoperation.

AREA OF EXPERTISE

- **Autonomous Systems & Robotics:** ROS/ROS 2, CARLA Simulator, MPC, MPPI, Stanley Controller, Pure Pursuit (PPC), Vehicle Dynamics, Real-Time Learning & Control, Sensor Fusion
- **Machine Learning & Perception:** Deep Learning (Transformers, CNNs), Reinforcement Learning (RL), Generative AI, YOLO, PyTorch, TensorFlow, Scikit-Learn, Computer Vision, 3D Point Clouds (PCD), 3DGS
- **Software & Embedded Systems:** C/C++, Python, SQL (PostgreSQL, SQLAlchemy, psycopg2), Linux/Ubuntu, Git, SBCs (NVIDIA Jetson, Raspberry Pi), Microcontrollers (STM32, ARM Cortex-M, ESP32, Arduino)

POSITIONS OF RESPONSIBILITY

- **Mentored 15+ undergraduate peers** in ROS and mobile robotics through hands-on Tinkerers' Lab workshops.
- **Lead Technical Liaison:** Directed multi-week enterprise deployments and live system troubleshooting at Cyient.
- **Designed vehicle subsystems for SASTRA Racing Team (BAJA)** and organized events for the Math Club.
- **Mobilized on-ground disaster relief** and coordinated emergency logistical distribution for Red Cross.

EXTRACURRICULAR & INTERESTS

- **Endurance Athletics:** Distance runner logging **over 280 km and 21+ hours** this year with a competitive **4:30/km pace**.
- **Team Sports & Adaptability:** Active participant in diverse field sports; quick to learn and adapt to new disciplines.
- **Strategy Games & Motorsport:** Competitive **Chess and Sudoku solver**; Formula 1 telemetry and Cricket enthusiast.